

Question	Marking Guidance	Mark	Comments
03.1	1. Volume of (stock) bacteria (culture); 2. Concentration of (stock) bacteria (culture); 3. Concentration of glucose OR Concentration of (respiratory) substrate; 4. Volume of ammonium chloride; 5. Time (bacteria/culture left to divide); 6. Concentration/volume of oxygen; 7. Concentration/volume of nitrogen;	2 max (2 x AO2)	Accept vol. for volume and conc. for concentration. 1 and 2. If neither credited accept number/mass of bacteria for 1 mark. 1 and 2. Ignore volume/concentration of liquid culture. 3. Accept concentration of ion/named ion but ignore ammonium chloride. Ignore 'same bacteria/species/type'. 1, 2, 3, 4, 6. and 7. Ignore amount. 3. Ignore 'sugar'. 3 and 4. Ignore nutrients. 6 and 7. Ignore availability/access/exposure.

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03.2	(For) 1. Nitrogenase activity decreases with increase in ammonium chloride (concentration); 2. Nitrogenase activity zero with high (concentration of) ammonium chloride OR Ammonium chloride remains (in medium) when nitrogenase activity zero; (Against) 3. Only used one species; 4. (Inhibition/results) may be due to chloride (ions)	3 max (3 x AO3)	1 and 2. Accept ammonia for ammonium chloride. 1. Accept negative correlation between nitrogenase activity and ammonium chloride (concentration). 2. Accept nitrogenase activity zero at 80/100/120/above 60 ($\mu\text{g cm}^3$ of ammonium chloride). 2. Accept 'stops' for zero. 3. Accept only <i>Azotobacter</i> /A. <i>chroococcum</i> / <i>chroococcum</i> / one nitrogen-fixing bacterium / one type/strain used.

	OR (Investigation) uses ammonium chloride not ammonia;		4. Reject 'chlorine'. Ignore reference to statistical tests.
Question	Marking Guidance	Mark	Comments
03.3	1. Less/no ATP/energy required/used OR More ATP/energy available; 2. ATP/energy can be used for growth/synthesis/replication OR Lower (rate of) respiration required OR ATP for phosphorylation;	2 (2 x AO2)	2. Accept ATP can be used for 'other reactions', 'movement', 'active transport' or correctly named reaction. 2. Ignore 'aerobic', 'anaerobic' in context of respiration. 2. Reject ATP/energy used for/in respiration. 2. Reject mitosis.